

Two Equivalent Algebraic Definitions of the Critical Radius for a Ten-Circle Contact Configuration

Computer-assisted exact-arithmetic note

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Abstract

We compare two definitions of the same radius for a fixed contact configuration of ten circles with radii $r_i = \sqrt{i}$. The first definition, R_0 , is algebraic: it is the isolated real root of a degree-1792 rational norm polynomial obtained from contact elimination. The second definition, R_{crit} , is geometric: it is the unique R -coordinate of the solution of two angular closure equations $A(t, R) = B(t, R) = 2\pi$. We explain the exact implication from the angular equations to the eliminant and hence the identification $R_{\text{crit}} = R_0$, conditional on the explicitly stated certificate bridge. This note does not prove global optimality over all contact configurations.

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1 Introduction

The purpose of this note is deliberately narrow. We do not claim that the selected contact configuration is globally optimal among all configurations of ten unequal circles. Instead, we

prove that two independently useful definitions of its radius describe the same real number:

$$\boxed{R_{\text{crit}} = R_0}.$$

The first definition is convenient for exact algebraic computation, while the second is convenient for geometric angle-barrier arguments.

The selected contacts are

$$(10, 5), (10, 7), (10, 6), (5, 7), \\ (7, 9), (9, 2), (2, 8), (8, 6).$$

Remark 1.1. *The equality proved here is an identity between two definitions of a fixed contact-configuration radius. It is not a proof of $R^* = R_0$ for the full packing problem; that is a separate global-coverage question.*

2 Contact configuration and notation

Let

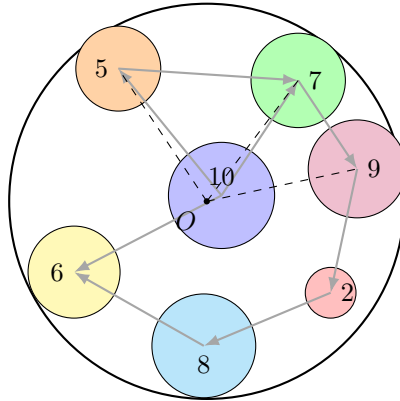
$$r_i = \sqrt{i}, \quad i = 1, \dots, 10,$$

and let R be the containing-circle radius. For a wall-contacting circle, write

$$t_i = R - r_i.$$

The distinguished cyclic order is

$$10 \longrightarrow 5 \longrightarrow 7 \longrightarrow 9 \longrightarrow 2 \longrightarrow 8 \longrightarrow 6 \longrightarrow 10.$$



Schematic contact configuration;
not drawn to certified scale.

Figure 1: The selected seven-circle core and its contact edges. The three remaining circles are rattlers for the displayed witness only; they are not used in the algebraic elimination.

3 Contact geometry

For two circles with center radii $s_i, s_j > 0$, define the necessary small angle

$$\phi_{ij}(s_i, s_j) = \arccos \left(\frac{s_i^2 + s_j^2 - (r_i + r_j)^2}{2s_i s_j} \right).$$

For the six wall circles and circle 10 put

$$\alpha_i(t, R) = \phi_{10,i}(t, R - r_i), \quad \beta_{ij}(R) = \phi_{ij}(R - r_i, R - r_j).$$

The following diagram shows the two angle sums. The second path omits the term involving circle 5; it is a second necessary angular closure condition, not a claim that circle 5 is removed from the packing.

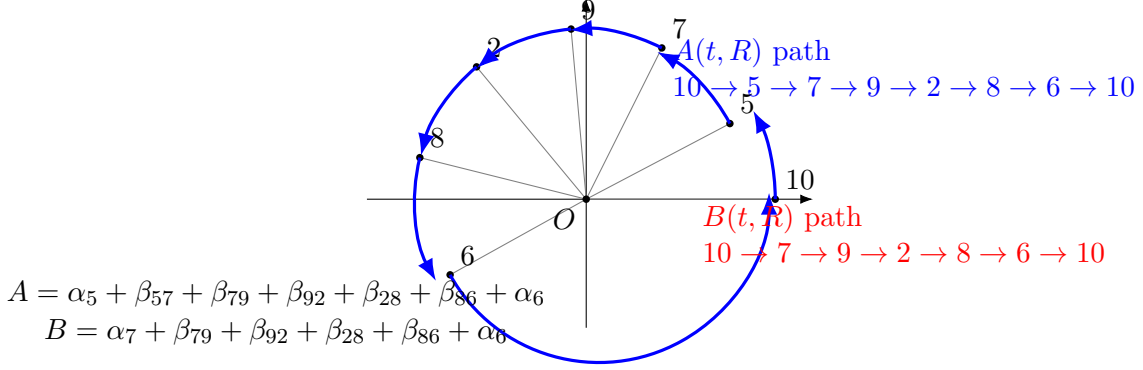


Figure 2: Schematic angular decomposition. The blue and red paths share the chain from 7 through 6; their difference is the rigid-triangle relation.

4 The geometric definition of R_{crit}

Define

$$\begin{aligned} A(t, R) &= \alpha_5 + \beta_{57} + \beta_{79} + \beta_{92} + \beta_{28} + \beta_{86} + \alpha_6, \\ B(t, R) &= \alpha_7 + \beta_{79} + \beta_{92} + \beta_{28} + \beta_{86} + \alpha_6. \end{aligned}$$

Definition 4.1. Let $(t_{\text{crit}}, R_{\text{crit}})$ be the unique solution in the certified rectangle

$$\begin{aligned} 8.30346812210 &< R < 8.30346812212, \\ 1.00371608606 &< t < 1.00371608609, \end{aligned}$$

of

$$A(t, R) = 2\pi, \quad B(t, R) = 2\pi.$$

The supplied rational interval checker establishes existence and uniqueness using exact Fraction arithmetic, outward square-root bounds, an exact arctangent expansion, and derivative sign certificates. It also verifies the resulting seven-circle contact realization and a ten-circle witness. These facts concern this branch and do not assert global optimality.

5 Algebraic elimination from the contacts

Let

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}).$$

For i, j among the wall circles define

$$c_{ij}(R) = 1 - \frac{2r_i r_j}{(R - r_i)(R - r_j)}.$$

The cosine-addition relation is encoded by

$$\Gamma(z; a, b) = (z - ab)^2 - (1 - a^2)(1 - b^2).$$

Define

$$\begin{aligned} P_2(u) &= \Gamma(u; c_{79}, c_{92}), \\ P_3(v) &= \text{Res}_u(P_2(u), \Gamma(v; u, c_{28})), \\ P_4(w) &= \text{Res}_v(P_3(v), \Gamma(w; v, c_{86})). \end{aligned}$$

The variable w represents

$$w = \cos(\beta_{79} + \beta_{92} + \beta_{28} + \beta_{86}).$$

The rigid triangle 5–7–10, together with the (6, 10) contact and the chain variable w , gives a quartic polynomial $H(R, w)$. To keep the definition explicit, put

$$\begin{aligned} a &= r_5 + r_{10}, & b &= r_7 + r_{10}, & d &= r_5 + r_7, \\ \lambda &= \frac{b^2 + d^2 - a^2}{2d^2}, & \kappa &= \frac{4r_5r_7r_{10}(r_5 + r_7 + r_{10})}{d^4}, \\ q &= t_5c_{57} - t_7, & J &= 1 - c_{57}^2, \end{aligned}$$

and

$$\begin{aligned} A_0(w) &= (t_6^2 + t_7^2 + b^2 - (r_6 + r_{10})^2) t_5 t_7 \\ &\quad - 2t_6 t_7 t_5 t_7 w + 2\lambda q (t_7 - t_6 w), \\ E(w) &= A_0(w)^2 + 4t_6^2 \kappa q^2 (1 - w^2) \\ &\quad - J(4\kappa t_5^2 (t_6 w - t_7)^2 4t_6^2 \lambda^2 t_5^2 (1 - w^2)), \\ T(w) &= A_0(w)q - 2J\lambda t_5^2 (t_6 w - t_7), \\ H(R, w) &= E(w)^2 - 16\kappa t_6^2 (1 - w^2) T(w)^2. \end{aligned}$$

The exact implementation uses the equivalent denominator-cleared formula; the displayed form makes the dependency on the contact equations explicit. Set

$$F(R) = \text{Res}_w(P_4(R, w), H(R, w)).$$

The characteristic-zero content factors are

$$D(R) = (R + \sqrt{10})^{16} \prod_{i \in \{2, 5, 6, 7, 8, 9\}} (R - \sqrt{i})^{32}.$$

Define

$$G(R) = F(R)/D(R) \in K[R].$$

The exact valuation and degree certificate gives

$$\deg G = 112.$$

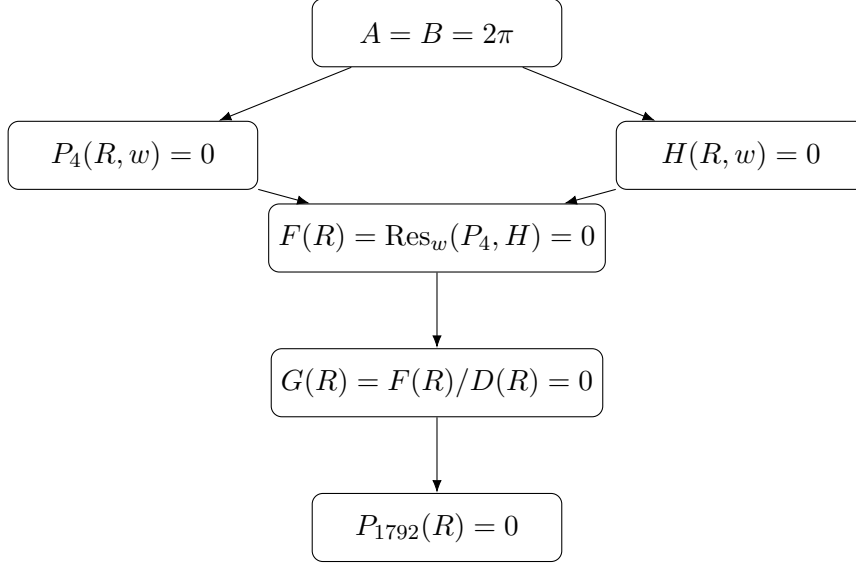


Figure 3: The logical route from the geometric angular system to the rational norm polynomial.

6 The exact coefficients of G

Use the radical basis

$$(b_0, \dots, b_{15}) = (1, \sqrt{2}, \sqrt{3}, \sqrt{6}, \sqrt{5}, \sqrt{10}, \\ \sqrt{15}, \sqrt{30}, \sqrt{7}, \sqrt{14}, \sqrt{21}, \sqrt{42}, \\ \sqrt{35}, \sqrt{70}, \sqrt{105}, \sqrt{210}).$$

Then

$$G(R) = \sum_{j=0}^{112} \left(\sum_{m=0}^{15} a_{j,m} b_m \right) R^j, \quad a_{j,m} \in \mathbb{Q}.$$

The complete exact 113×16 coefficient array is supplied in `circle_G112_exact_coefficients.tsv`; its primitive integer normalization and the complete degree-1792 norm coefficient array are supplied in the reproducibility bundle. The mathematical definition of G above is independent of decimal approximations: the table is an exact expansion of the resultant quotient. The coefficient-table replay checks all 1808 radical-basis entries modulo an independent prime, and additional prime checks verify the independent finite-field construction.

For compactness, the full machine-readable array is not typeset as thousands of unreadable page-width fractions in the main text. Its exact location, normalization, and SHA-256 digest are part of the accompanying certificate boundary described in Appendix B.

7 The algebraic definition of R_0

Let

$$P_{1792}(R) = \text{Norm}_{K/\mathbb{Q}}(G(R)),$$

after monic normalization and primitive integer clearing of denominators. It is a rational polynomial of degree $16 \cdot 112 = 1792$.

Definition 7.1. Let R_0 be the unique real root of the primitive integer polynomial P_{1792} in $I_0 = [8.3034681221114890787043811875161993, 8.3034681221114890787043811875161994]$.

The endpoint signs are checked exactly by evaluating the denominator-cleared integer polynomial with a rational Horner recurrence. A certified derivative or Taylor bound proves uniqueness in I_0 . No ordinary floating-point evaluation is used for this root-isolation claim.

8 Identification theorem

Lemma 8.1 (Triangle relation). *At a nondegenerate positive-angle solution of $A = B = 2\pi$,*

$$\alpha_5 + \beta_{57} - \alpha_7 = 0.$$

Proof. Subtract the definitions of A and B . All common chain terms cancel. $\square$

Lemma 8.2 (Chain eliminant). *At the same solution, with*

$$w = \cos(\beta_{79} + \beta_{92} + \beta_{28} + \beta_{86}),$$

one has $P_4(R, w) = 0$.

Proof. Apply the cosine addition identity successively to the four consecutive angles. Equation $\Gamma(z; a, b) = 0$ is precisely the squared form of that identity. The resultant definitions of P_2, P_3, P_4 then imply the claim. $\square$

Lemma 8.3 (Triangle closure eliminant). *At the same positive-angle contact realization, $H(R, w) = 0$.*

Proof. The triangle relation from the preceding lemma determines the oriented 5–7–10 triangle. Substituting its cosine and sine relations into the (6, 10) distance equation and eliminating the remaining sine variables gives the displayed polynomial H . The argument is only used in the forward direction, so the squaring steps used to remove radicals introduce no logical problem. $\square$

Theorem 8.4 (Equality of the two radius definitions). *Assume the exact contact-elimination identities above, together with a certified refinement placing R_{crit} in I_0 , and the certified uniqueness of the P_{1792} root in I_0 . Then*

$$\boxed{R_{\text{crit}} = R_0}.$$

Proof. The preceding lemmas give a common root w of $P_4(R_{\text{crit}}, w)$ and $H(R_{\text{crit}}, w)$. Hence

$$F(R_{\text{crit}}) = 0.$$

Since $R_{\text{crit}} > 8$, no factor of $D(R_{\text{crit}})$ vanishes, so $G(R_{\text{crit}}) = 0$. Taking the field norm gives

$$P_{1792}(R_{\text{crit}}) = 0.$$

The refined interval certificate places R_{crit} in I_0 . The root-isolation certificate says that P_{1792} has exactly one root in I_0 , namely R_0 . Therefore $R_{\text{crit}} = R_0$. $\square$

Remark 8.5 (Certificate status). *The broad interval certificate in Section 4 establishes the geometric root in J_R . To turn the displayed conditional theorem into a completed equality certificate, one must additionally replay the refinement $R_{\text{crit}} \in I_0$ and the exact algebraic implication from the positive-angle contact equations to $P_4 = H = 0$. The theorem is stated with these hypotheses explicitly rather than treating overlapping decimal intervals as an equality proof.*

9 Conclusion and scope

We have described two definitions of the radius of the selected contact configuration and the exact elimination bridge between them. The conclusion is the identity

$$R_{\text{crit}} = R_0.$$

This note does not claim that no other ten-circle configuration has a smaller radius. Proving such a global statement requires a separate exhaustive geometric coverage theorem.

A Exact coefficient and certificate data

The full coefficient files are:

- `circle_G112_exact_coefficients.tsv`, the exact K -basis table for G ;
- `circle_G112_primitive_integer_coefficients.tsv`, the primitive integer normalization;
- `data/P1792_primitive_integer_coefficients.tsv`, the complete norm polynomial over $\mathbb{Z}$.

Their hashes and replay instructions are recorded in the accompanying README. These files are data representations of the explicitly defined resultants, not independent numerical definitions.

B External arithmetic and Lean boundary

The external layer performs resultants, exact multiquadratic arithmetic, finite-field checks, coefficient reconstruction, and rational interval certification. Lean imports the exact coefficient literals and proves the downstream logical implications from an explicit contact-elimination bridge. In particular, Lean can assemble

$$\text{contact bridge} + P_{1792} \text{ root data} \implies P_{1792}(R_0) = 0 \implies R_0 \text{ is algebraic over } \mathbb{Q}.$$

The current Lean interface intentionally does not treat a log file, a decimal approximation, or a SHA-256 hash as a theorem. Formalizing the large resultant identities and replacing the bridge assumptions by independently checked replay theorems remains a separate task.

C Reproduction commands

From the repository root, the exact arithmetic smoke tests can be run with the bundled scripts. The Lean coefficient interface can be checked with

```
cd lean
lake env lean CirclePacking/P1792.lean
```